

Optiver 



Trading options as a Market Maker



Optiver Trading Academy 2026



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Agenda

- What is an option?
- Forward pricing
- Option pricing
- Volatility
- Skew
- Delta



What is an option?

An option is a **right**, not an obligation
to buy (**call**) or to sell (**put**) an asset
at a certain price (**exercise price or strike**)
at a certain time in the future (**expiration**)



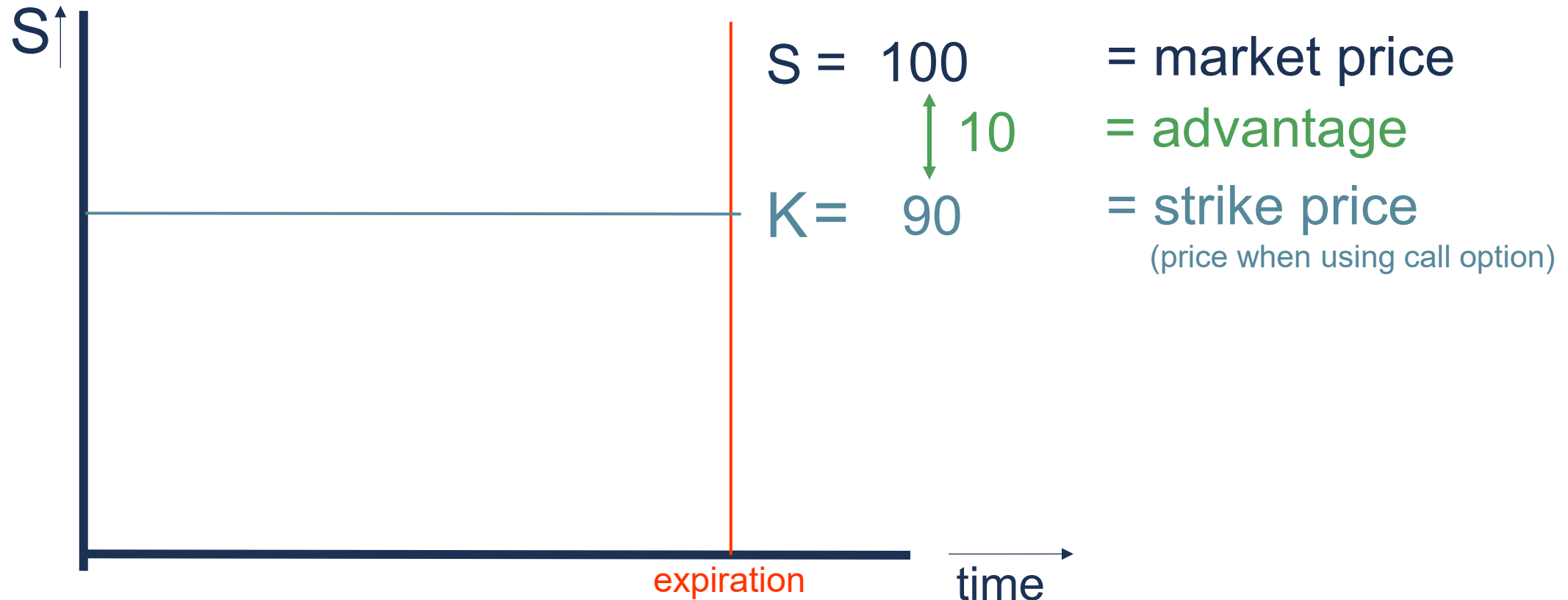
Assumptions

In this lecture we focus on:

- △ European style options
- △ Call options
- △ No dividends



Call option at expiration





Payoff of Call option

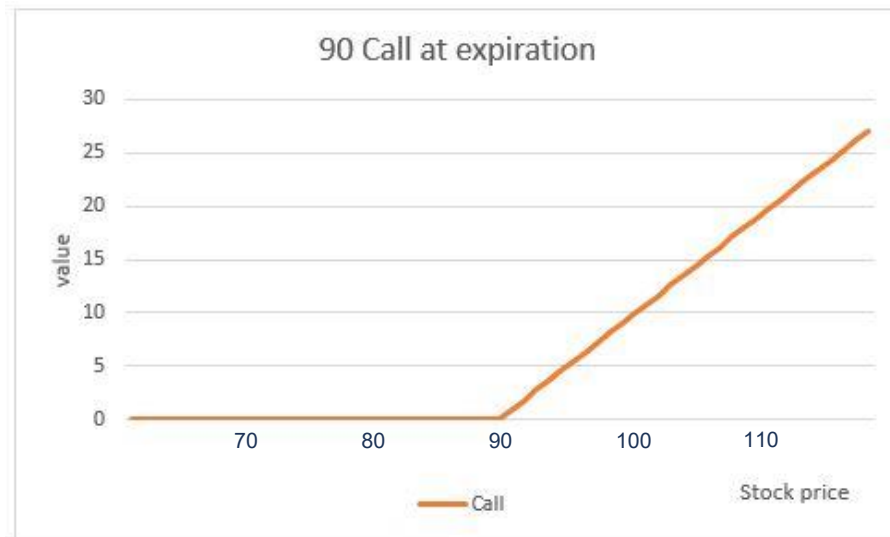
At expiration

$$C = \max [0, (S - K)]$$

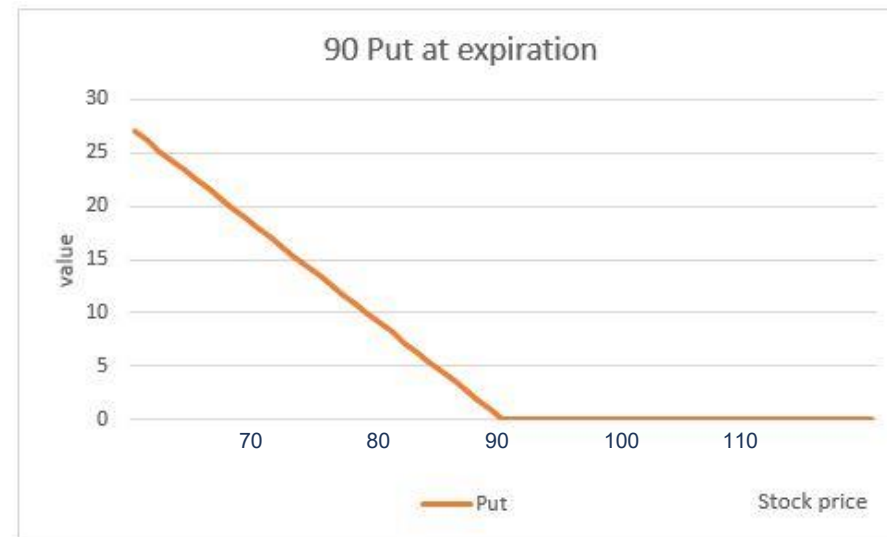


Expiration diagrams

Long call option



Long put option





Before expiration

What is the value of an option now?

In addition to any **intrinsic value**, there may also be a **time value**

Why would someone pay more for an option than just the intrinsic value?

How much more?

How do you determine this value?



Building blocks of an option

intrinsic
value

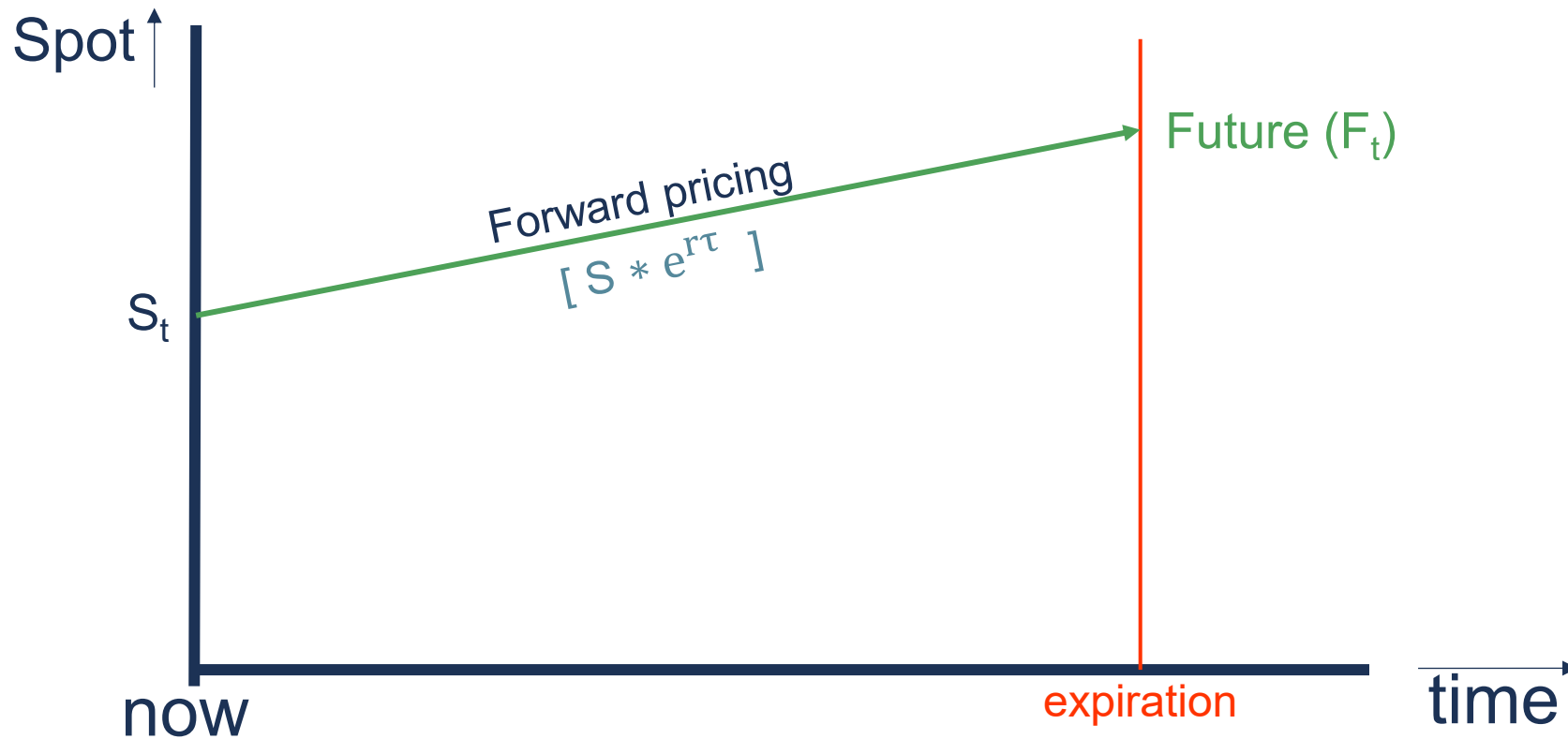
Intrinsic Value

time
value

Time Value (or Extrinsic value)

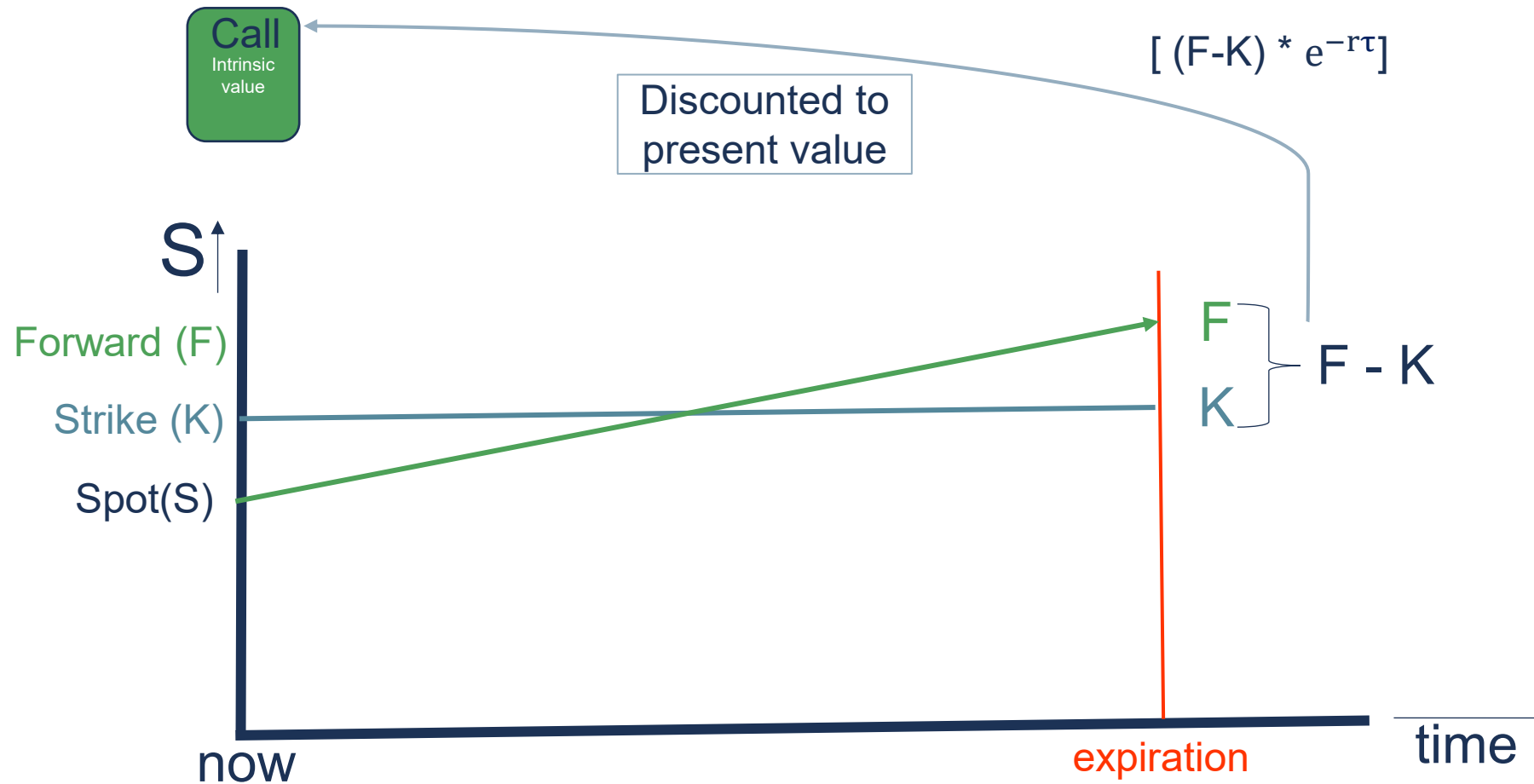
Future value

Note: we assume here that stocks do not pay dividends!



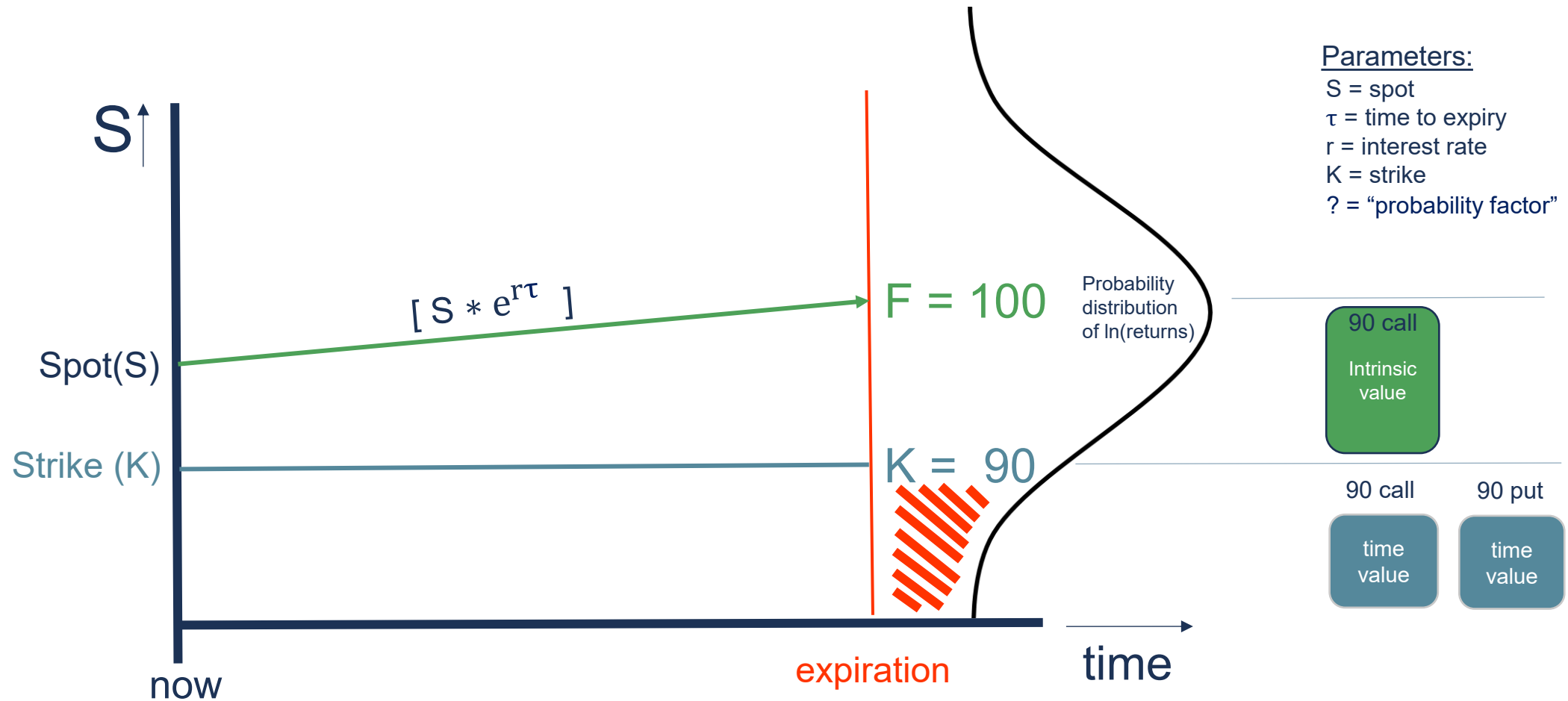


Intrinsic value





Time value





Black-Scholes

At expiration:

$$C = \max[0, (S - K)]$$

Before expiration:

$$C = S * N(d_1) - K * e^{-r\tau} * N(d_2)$$



Black-Scholes

Before Expiration

$$C = S * N(d_1) - K * e^{-r\tau} * N(d_2)$$

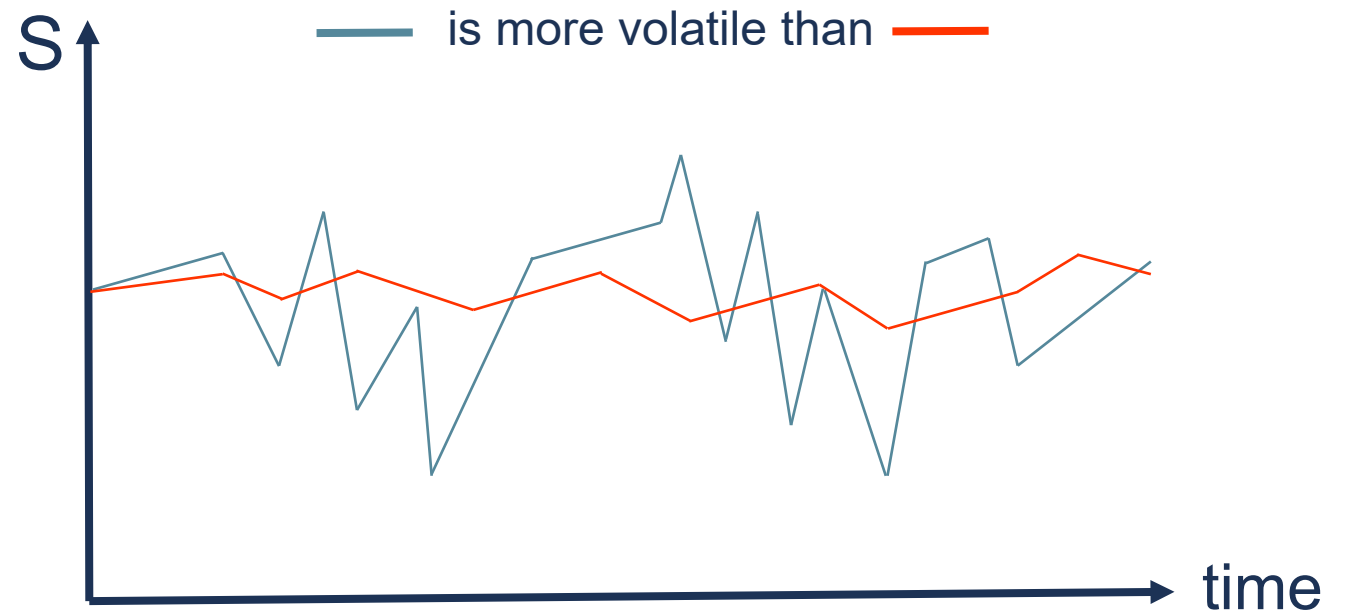
$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}}$$

$$d_2 = d_1 - \sigma\sqrt{\tau}$$



Historical Volatility

Based on the known price data from the past, the standard deviation of the price changes can be calculated. This standard deviation is called the **historical volatility**.



Implied Volatility





Pricing Parameters

S	spot = price underlying asset
K	exercise price = strike
τ	time to expiration
r	interest rate
σ	implied volatility

$$C = S * N(d_1) - K * e^{-rt} * N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}}$$

$$d_2 = d_1 - \sigma\sqrt{\tau}$$

Higher implied volatility $\longrightarrow$ higher prices for both calls and puts



In Practice...

Let's calculate the call prices for different strikes (**K**) and fill in the parameters below:

S spot = price underlying asset

τ time to expiration

r interest rate

σ implied volatility

$$C = S * N(d_1) - \mathbf{K} * e^{-r\tau} * N(d_2)$$

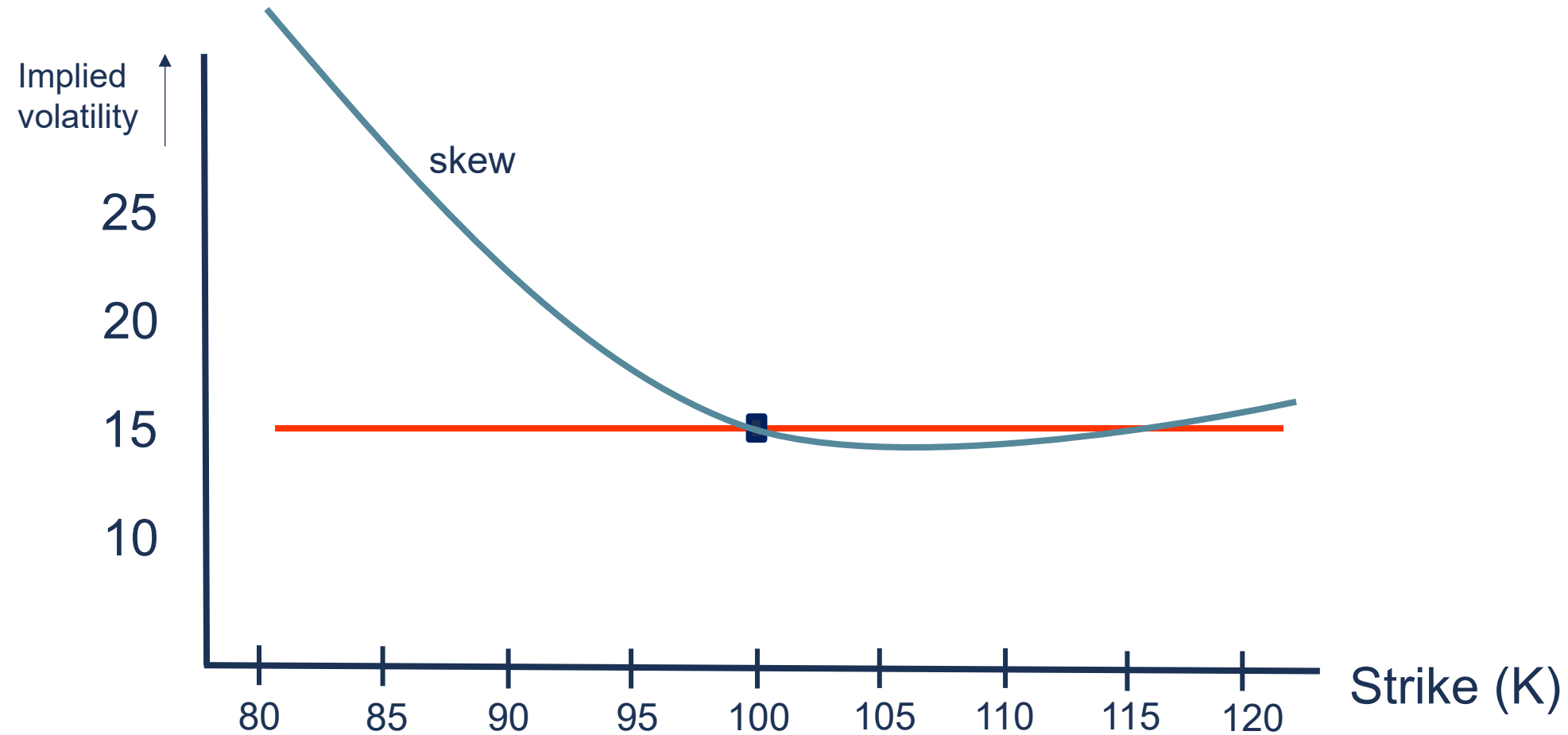
$$d_1 = \frac{\ln\left(\frac{S}{\mathbf{K}}\right) + \left(r + \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}}$$

$$d_2 = d_1 - \sigma\sqrt{\tau}$$

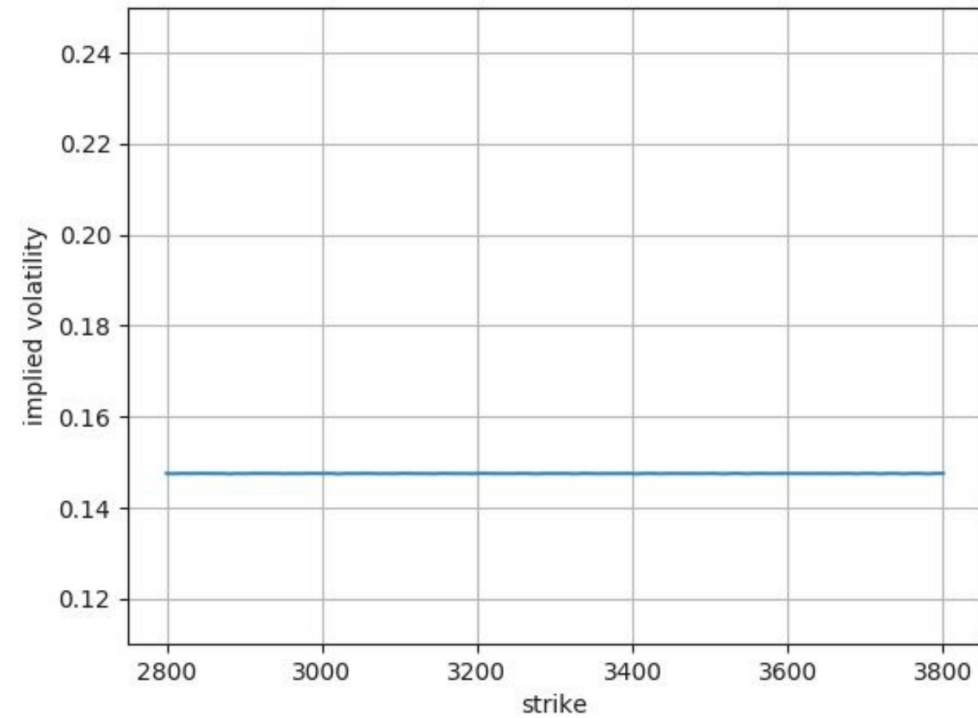
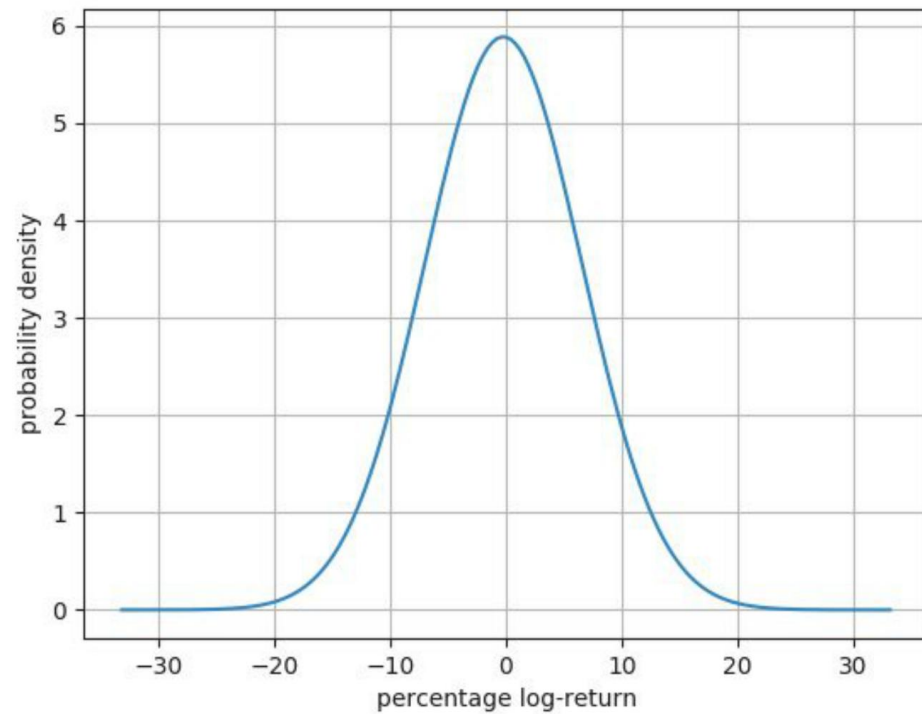


Implied Volatility

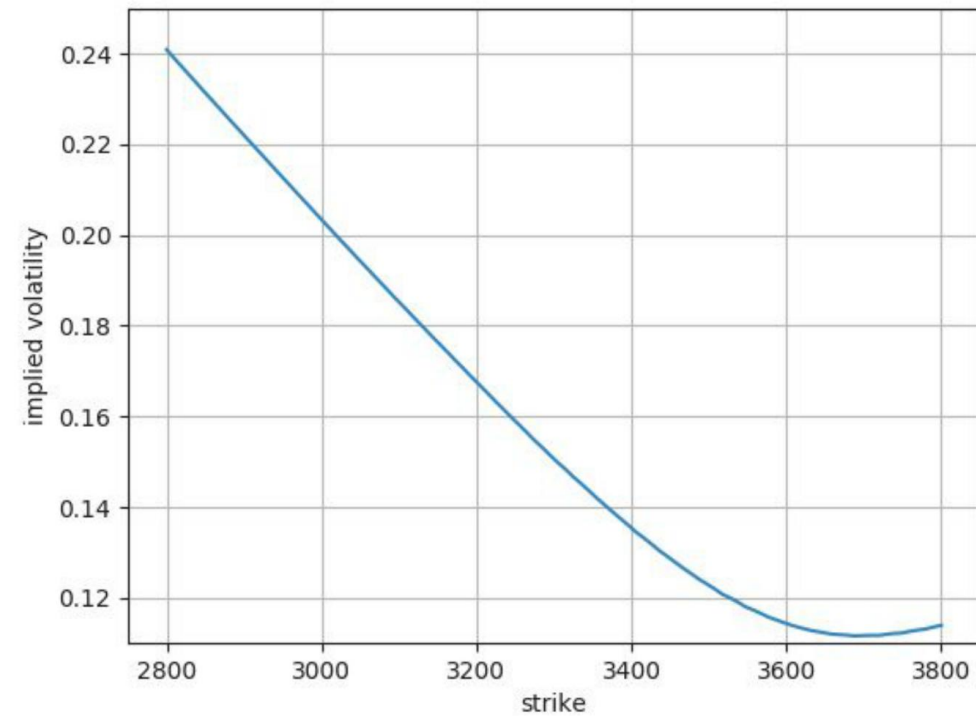
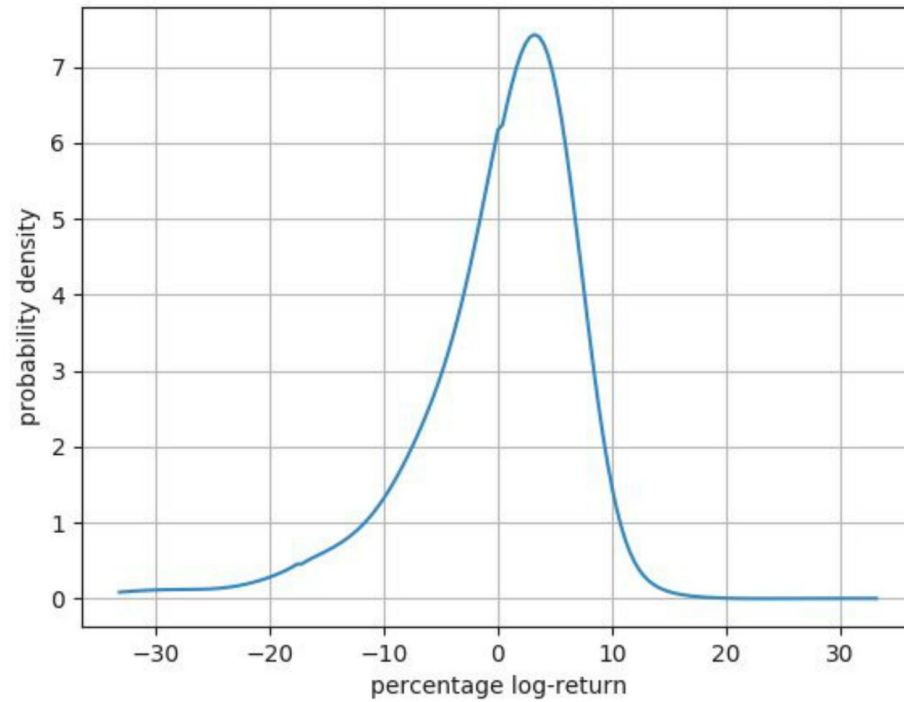
(for different strikes)



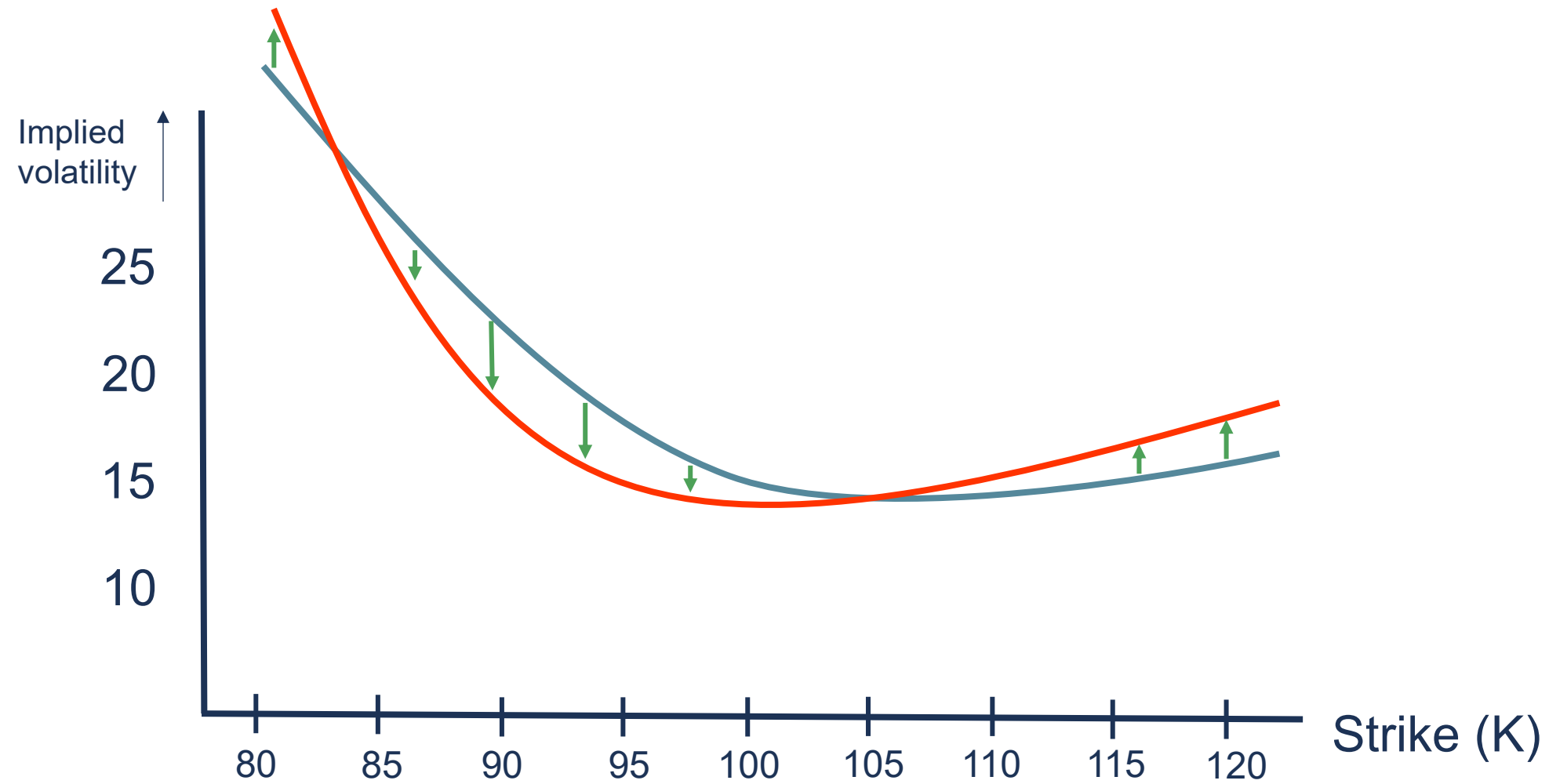
Normal Distribution



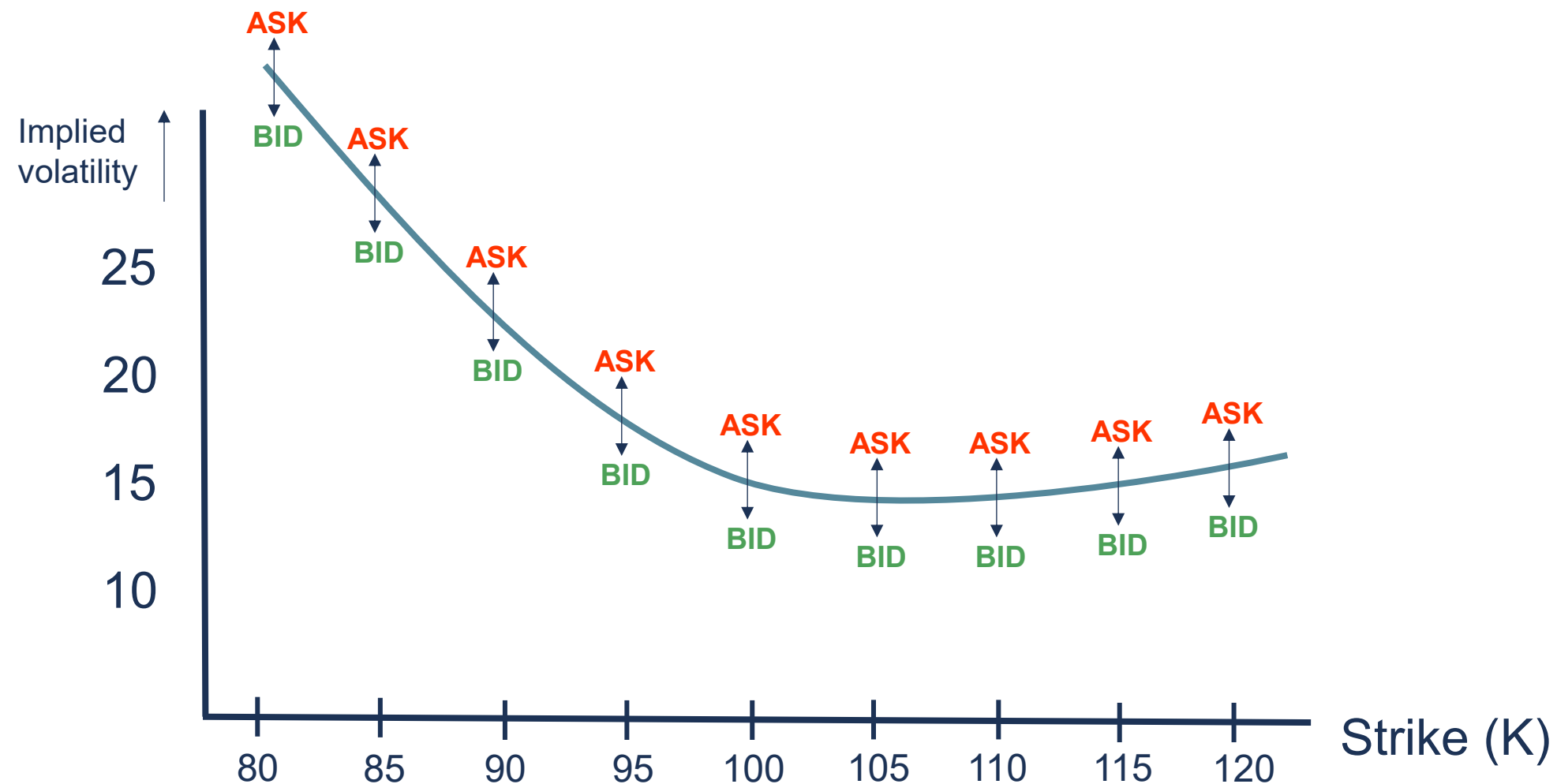
Skewed Distribution



Volatility Changes



Trading Volatility





What does a market maker do?

Based on the parameters:

 **S** spot = price underlying asset
K exercise price = strike
 τ time to expiration
r interest rate
 σ implied volatility

the MM calculates the call and put values and shows both bid and ask prices to the market

ask ask
| |
call value put value
| |
bid bid

and waits until someone else trades with the MM, i.e. buy/sell a call/put

buy call or
sell call or
buy put or
sell put



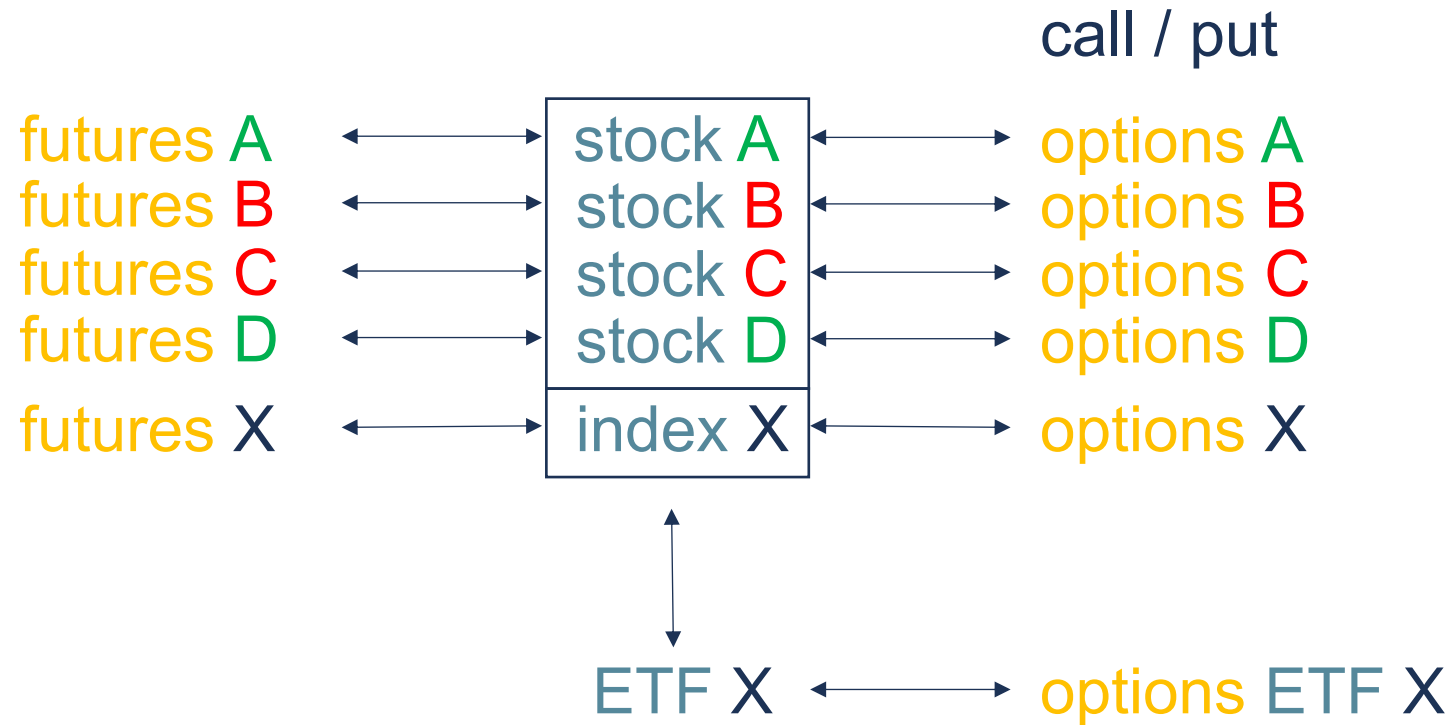
Delta

$$\Delta = \frac{\partial V}{\partial S} = \frac{\text{change in option value}}{\text{change in value of underlying asset}}$$

Delta interpretations

- 1) Change of value of an option (or portfolio) due to a change in the underlying
- 2) Probability of an option to expire in-the-money, i.e. with intrinsic value (by approximation)
- 3) The number of shares of the underlying asset in order to hedge, i.e. the equivalent weight of the underlying asset compared to the option

Financial Instruments





Market Making and Hedging

stock A

bid volume	price	ask volume
	44.67	8539
	44.66	10551
	44.65	3677
	44.64	6800
	44.63	9061
	44.62	6693
	44.61	6573
2049	44.60	
5176	44.59	
1080	44.58	
5000	44.57	
16776	44.56	
2000	44.55	
15349	44.54	
7800	44.53	
11327	44.52	

Stocks exchange PQR

1

price of stock A is input for
calculation of option price

low-latency

after a trade in option A
a hedge order in stock A
needs to be executed

2

option A

bid #	price	ask #
	24.95	20
	24.90	37
	24.85	10
	24.80	
	24.75	100
	24.70	
100	24.65	
2	24.60	
	24.55	
10	24.50	
6	24.45	
20	24.40	

Options exchange XYZ



Summary

- A market maker calculates option prices based on available market data
- The MM shows a bid and an offer to the market
- If the MM does a trade, (s)he hedges with the underlying asset (S) in order to be neutral for changes in the underlying asset (S)
- This hedge is done by using the delta of the option
- Effectively, a MM trades volatility since the other parameters are 'neutralized'
- Basically, a MM tries to buy options at a relatively low volatility and sell options at a relatively high volatility level



Thank you!



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